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328813 (28)

BE (8th Semester)

Examination, April-May, 2014

Branch : Et & T

INDUSTRIAL & POWER ELECTRONICS

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (i) Part (a) of each question is compulsory.

(ii) Solve any two parts from (b), (c) and (d).

Q. 1. (a) Distinguish between load regulation and source regulation ? 2

(b) Design a 10 V zener voltage regulator for the following conditions : 7

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(i) The load current ranges from 50 mA to 100 mA.

(ii) The source (input) voltage ranges from 12 V to 15 V.

(c) What purpose the IC LM340 serves ? Explain in details. 7

(d) How does the linear power supply differs from SMPS ? 7

Q. 2. (a) What is the sufficient condition for turning OFF a SCR ? 2

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(b) A SCR has $V_G - i_G$ relation as $V_G = 1 + 10\, i_G$. In an application in which gating voltage of 20 volt rectangular pulses of 180° duration is applied. Calculate the resistance R_G to be connected in series with the gate voltage source in order to limit average gate power loss to be 0.5 W. Find the values of V_G and i_G during gating. 7

(c) Explain resistance firing circuit (anode triggering) for firing of SCR ? 7

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(d) Explain the necessity of static equalizing network for three SCRs connected in series ? 7

Q. 3. (a) What do you mean by chopper and where it is used ? 2

(b) Explain the working principle of single phase dual converter ? 7

(c) Explain single phase mid-point cycloconverter with the help of circuit diagram ? 7

(d) Describe McMurray bridge inverter in brief ? 7

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Q. 4. (a) What is skin effect ? 2

(b) Explain the principle of high frequency dielectric per unit volume ? 7

(c) Explain the applications and merits & demerits of induction heating ? 7

(d) A metal cylinder 5 cm long and 1 cm diameter is placed inside a coil having 10 turns. The coil carries current of 100 A. If the depth of heating is 0.4 mm determine the frequency and also heat generated per unit surface area. Assume $\mu_r = 1$ and resistivity of the metal to be 5×10^{-8} Ω-m. 7

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Q. 5. (a) What do you mean by A.C. power conditioning ? 2

(b) What are the different types of noises ? Explain in brief ? 7

(c) What is the need of UPS ? Explain it with proper block diagram. 7

(d) Write a short note on constant voltage transformer ? 7

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